

## PH 611: Nuclear and Particle Physics

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|---------------|-----------------------------------------------------------|
| Course Code   | PH 611                                                    |
| Course Name   | Nuclear and Particle Physics                              |
| L-T-P-C       | 4-0-0-4                                                   |
| Prerequisites | PH 514 Quantum Mechanics-I, PH 521 Electromagnetic Theory |
| Intended for  | UG/PG                                                     |
| Distribution  | Core for I-Ph.D.; Elective for the rest                   |
| Approval      | 9th Senate (May 2016)                                     |

### Course Contents

- **Properties of Nuclei**
  - Nuclear size, Rutherford scattering, nuclear radius and charge distribution, nuclear form factor, mass and binding energy, semiempirical mass formula, angular momentum, parity and isospin, magnetic dipole moment electric quadrupole moment and nuclear shape, experimental determination. [4 Lectures]
- **Two-body problems**
  - Deuteron ground state, excited states, spin dependence of nuclear forces, electromagnetic moment and magnetic dipole moment of deuteron and the necessity of tensor forces, two nucleon scattering, n-p scattering, partial wave analysis, phase-shift, scattering length, p-p scattering, charge symmetry and charge independence of nuclear forces, exchange nature of nuclear forces, Yukawa's theory. [6 Lectures]
- **Nuclear decay**
  - Beta emission and electron capture, Fermi's theory of beta decay, selecti Liquid drop model, Fermi gas model, shell and collective model. [5 Lectures]
- **Nuclear models**
  - Liquid drop model, Fermi gas model, shell and collective model. [5 Lectures]
- **Nuclear Reactions**
  - Different types of reactions, conservation laws, energetics, isospin, reaction crosssection, resonance scattering and reactions, Breit-Wigner dispersion relation, Compound nucleus formation and break-up, Optical model, transfer reactions, nuclear fission, neutron physics, fusion reaction. [7 Lectures]
- **Particle accelerators and detectors**
  - Electrostatic accelerators, cyclotron, Synchrotron and synchroscyclotron, linear accelerators, colliding beam accelerators, ionization chamber, scintillation detectors, semiconductor detectors. [5 Lectures]
- **Elementary particles**
  - Fundamental interactions, properties mesons and baryons, symmetries and conservation laws, charge-conjugation, parity and time reversal, CPT theorem, Gell-Mann-Nishijima formula, intrinsic parity of pions, resonances, symmetry classification of elementary particles, quark model, concept of colour charge, discrete symmetries, properties of quarks and leptons, gauge symmetry in electrodynamics, particle interactions and Feynman diagrams. [8 Lectures]

### Text Books

1. K. S. Krane, **Introductory Nuclear Physics**, John Wiley.

### References

1. W. E. Burcham and M. Jobes, **Nuclear and particle Physics**, John Wiley & Sons.
2. D. J. Griffiths, **Introduction to Elementary Particles**, John Wiley & Sons.
3. A. Das and T. Ferbel, **Introduction to nuclear and particle physics**, John Wiley.
4. M. A. Preston and R. K. Bhaduri, **Structure of the nucleus**, Addison-Wesley.
5. S. N. Ghoshal, **Atomic and Nuclear Physics**, Vol. 2.
6. Roy and B. P. Nigam, **Nuclear Physics: Theory and Experiment**, New Age.
7. D. Perkins, **Introduction to High Energy Physics**, 4th Edition, Cambridge University Press, 2000.